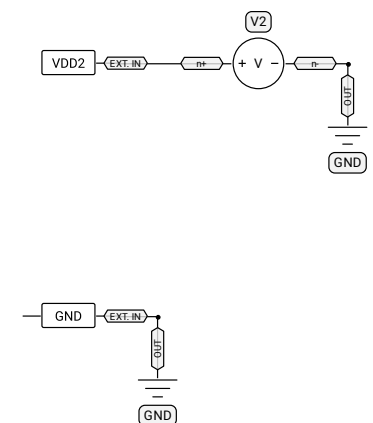




VDD (PIN 20)	
Property	Value
Min. value (V)	2.30
Typ. value (V)	5.00
Max. value (V)	5.50

PIN 19 (IO0) Label: "LOAD"	
Property	Value
I/O selection	Digital input
In mode	Digital in without Schmitt trigger
Out mode	None
Resistor	Floating

PIN 18 (IO1) Label: "CAL_MODE"	
Property	Value
I/O selection	Digital input
In mode	Digital in without Schmitt trigger
Out mode	None
Resistor	Pull Down
Resistor value	1M

PIN 17 (IO2) Label: "SAVE"	
Property	Value
I/O selection	Digital input
In mode	Digital in without Schmitt trigger
Out mode	None
Resistor	Floating

PIN 16 (IO3) Label: "GAIN_X0"	
Property	Value
I/O selection	Digital input
In mode	Digital in without Schmitt trigger
Out mode	None
Resistor	Floating

PIN 15 (IO4) Label: "GAIN_0X"	
Property	Value
I/O selection	Digital input
In mode	Digital in without Schmitt trigger
Out mode	None
Resistor	Floating

PIN 14 (IO5)	
Property	Value
I/O selection	Digital input
In mode	Digital in without Schmitt trigger
Out mode	None
Resistor	Floating

PIN 13 (SCL) Label: "SCL"	
Property	Value
I/O selection	Digital input
In mode	Digital in without Schmitt trigger
Out mode	None

PIN 12 (SDA) Label: "SDA"	
Property	Value
I/O selection	Digital input
In mode	Digital in without Schmitt trigger
Out mode	None

PIN 11 (IO6)	
Property	Value
I/O selection	Digital output
In mode	None
Out mode	2x push pull

PIN 9 (IO7) Label: "GAIN_0"	
Property	Value
I/O selection	Digital output
In mode	None
Out mode	1x push pull

PIN 8 (IO8) Label: "GAIN_1"	
Property	Value
I/O selection	Digital output
In mode	None
Out mode	1x push pull

VDD2 (PIN 7)	
Property	Value
Min. value (V)	1.71
Typ. value (V)	5.00
Max. value (V)	5.50

PIN 6 (IO9) Label: "GAIN_2"	
Property	Value
I/O selection	Digital output
In mode	None
Out mode	1x push pull

PIN 5 (IO10) Label: "GAIN_3"	
Property	Value
I/O selection	Digital output
In mode	None
Out mode	1x push pull

PIN 4 (IO11) Label: "LOAD_G"	
<i>Property</i>	<i>Value</i>
I/O selection	Digital output
In mode	None
Out mode	1x push pull

PIN 3 (IO12) Label: "OTP"	
<i>Property</i>	<i>Value</i>
I/O selection	Analog input/output
In mode	Analog input/output
Out mode	Analog input/output
Resistor	Floating

PIN 2 (IO13) Label: "OCP"	
<i>Property</i>	<i>Value</i>
I/O selection	Analog input/output
In mode	Analog input/output
Out mode	Analog input/output
Resistor	Pull Down
Resistor value	1M

PIN 1 (IO14) Label: "OVP"	
<i>Property</i>	<i>Value</i>
I/O selection	Analog input/output
In mode	Analog input/output
Out mode	Analog input/output
Resistor	Pull Down
Resistor value	1M
100uA pullup on input	Enable

A CMP0H	
<i>Property</i>	<i>Value</i>
100uA pullup on input	Enable
Hysteresis	Disable
IN+ gain	Disable
IN+ source	PIN 1 (IO14)
IN- source	1024 mV

A CMP1H	
<i>Property</i>	<i>Value</i>
Hysteresis	Disable
IN+ gain	Disable
IN+ source	PIN 2 (IO13)
IN- source	1024 mV

A CMP2L	
<i>Property</i>	<i>Value</i>
Hysteresis	Disable
IN+ gain	Disable
IN+ source	PIN 3 (IO12)
IN- source	1920 mV

2-bit LUT0/DFF/LATCH0			
<i>IN1</i>	<i>IN0</i>	<i>OUT</i>	
0	0	0	
0	1	1	
1	0	1	
1	1	1	
<i>Property</i>		<i>Value</i>	
Type		LUT	
Standard gates		OR	

2-bit LUT1/DFF/LATCH1		
<i>IN1</i>	<i>IN0</i>	<i>OUT</i>
0	0	1
0	1	0
1	0	0
1	1	0
<i>Property</i>		<i>Value</i>
Type		LUT
Standard gates		NOR

2-bit LUT3/PGEN		
<i>IN1</i>	<i>IN0</i>	<i>OUT</i>
0	0	0
0	1	0
1	0	0
1	1	1
<i>Property</i>		<i>Value</i>
Type		LUT
Standard gates		AND

3-bit LUT0/DFF/LATCH3			
<i>IN2</i>	<i>IN1</i>	<i>IN0</i>	<i>OUT</i>
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	0
<i>Property</i>		<i>Value</i>	
Type		LUT	
Standard gates		Defined by user	

3-bit LUT1/DFF/LATCH4			
<i>IN2</i>	<i>IN1</i>	<i>IN0</i>	<i>OUT</i>
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	0
<i>Property</i>		<i>Value</i>	
Type		LUT	
Standard gates		NOR	

3-bit LUT2/DFF/LATCH5			
IN2	IN1	IN0	OUT
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	0

Property	Value
Type	LUT
Standard gates	Defined by user

3-bit LUT3/DFF/LATCH6			
<i>IN2</i>	<i>IN1</i>	<i>IN0</i>	<i>OUT</i>
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1
<i>Property</i>		<i>Value</i>	
Type		LUT	
Standard gates		Defined by user	

3-bit LUT4/DFF/LATCH7	
Property	Value
Type	DFF / LATCH
Mode	DFF
nSET/nRESET option	nRESET
Initial polarity	Low
Q output polarity	Non-inverted (Q)

3-bit LUT5/DFF/LATCH8	
Property	Value
Type	DFF / LATCH
Mode	DFF
nSET/nRESET option	nRESET
Initial polarity	Low
Q output polarity	Non-inverted (Q)

MF0 (4-bit LUT0, DFF/LATCH9, WS Ctrl, 16-bit CNT0/DLY0/FSM0)				
4-bit LUT0 (MF0)				
IN3	IN2	IN1	IN0	OUT
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	0
0	1	1	1	0
1	0	0	0	1
1	0	0	1	0
1	0	1	0	0
1	0	1	1	0
1	1	0	0	0
1	1	0	1	0
1	1	1	0	0
1	1	1	1	0
Property		Value		
Standard gates		Defined by user		

I2C				
Property	Value			
IO Latching	Enable			
Mode selection	Standard/fast mode			
Control code, bin	0001			
Control byte, read/ write	0x11 / 0x10			
Device address, dec/ hex	8 / 0x08			
PIN 19 (IO0) output expander select	From matrix			
PIN 14 (IO5) output expander select	From matrix			
PIN 11 (IO6) output expander select	From matrix			
PIN 6 (IO9) output expander select	From matrix			
Virtual OUT0	0			
Virtual OUT1	0			
Virtual OUT2	0			
Virtual OUT3	0			
Virtual OUT4	0			
Virtual OUT5	0			
Virtual OUT6	0			
Virtual OUT7	0			
PIN 19 (IO0) OUT	0			
PIN 14 (IO5) OUT	0			
PIN 11 (IO6) OUT	0			
PIN 6 (IO9) OUT	0			
Control code selection	#3	#2	#1	#0
	0	0	0	1

External Components

V1	
Property	Value
Element	Voltage Source
Pre-start delay	0s
Internal capacitance	100nF
Internal resistance	100hm
Pre-start state	Low
Type	DC
DC Voltage	3.3V
Ramp rise time	1ms

V2	
Property	Value
Element	Voltage Source
Pre-start delay	0s
Internal capacitance	100nF
Internal resistance	100hm
Pre-start state	Low
Type	DC
DC Voltage	3.3V
Ramp rise time	1ms

V3	
Property	Value
Element	Voltage Source
Type	DC
DC Voltage	3.3V

Project Specs			
	<i>Min.</i>	<i>Typ.</i>	<i>Max.</i>
VDD	2.30	5.00	5.50
VDD2 (V):	1.71	5.00	5.50
Temperature (°C):	-40.00	25.00	85.00

General Settings	
GPIO quick charge	Disable
Pattern ID	1
2k Register lock status	Unlocked
2k NVM lock status	Unlocked
Emulated EEPROM lock status	Unlocked
Protect entire lock configuration	Disable

I2C Probing

Probe pins			
Pin	Signal name	Register	In use
GND (PIN 10) -> OUT		HEX: 0x74`b0 DEC: 116`b0	+
PIN 19 (IO0) -> OUT	NET21	HEX: 0x74`b1 DEC: 116`b1	+
PIN 18 (IO1) -> OUT	NET12	HEX: 0x74`b2 DEC: 116`b2	+
PIN 17 (IO2) -> OUT	NET25	HEX: 0x74`b3 DEC: 116`b3	+
PIN 16 (IO3) -> OUT	NET18	HEX: 0x74`b4 DEC: 116`b4	+
PIN 15 (IO4) -> OUT	NET26	HEX: 0x74`b5 DEC: 116`b5	+
PIN 14 (IO5) -> OUT		HEX: 0x74`b6 DEC: 116`b6	-
PIN 8 (IO8) -> OUT		HEX: 0x74`b7 DEC: 116`b7	-
PIN 6 (IO9) -> OUT		HEX: 0x75`b0 DEC: 117`b0	-
PIN 5 (IO10) -> OUT		HEX: 0x75`b1 DEC: 117`b1	-
PIN 4 (IO11) -> OUT		HEX: 0x75`b2 DEC: 117`b2	-
PIN 3 (IO12) -> OUT	NET10	HEX: 0x75`b3 DEC: 117`b3	-
PIN 2 (IO13) -> OUT	NET9	HEX: 0x75`b4 DEC: 117`b4	-
PIN 1 (IO14) -> OUT	NET8	HEX: 0x75`b5 DEC: 117`b5	-
2-bit LUT0/DFF/ LATCH0 -> OUT	NET22	HEX: 0x75`b6 DEC: 117`b6	+
2-bit LUT1/DFF/ LATCH1 -> OUT	NET11	HEX: 0x75`b7 DEC: 117`b7	+
2-bit LUT2/DFF/ LATCH2 -> OUT		HEX: 0x76`b0 DEC: 118`b0	-
2-bit LUT3/PGEN ->	NET24	HEX: 0x76`b1	+

Probe pins			
OUT		DEC: 118`b1	
3-bit LUT0/DFF/ LATCH3 -> OUT	NET16	HEX: 0x76`b2 DEC: 118`b2	+
3-bit LUT1/DFF/ LATCH4 -> OUT	NET20	HEX: 0x76`b3 DEC: 118`b3	+
3-bit LUT2/DFF/ LATCH5 -> OUT	NET17	HEX: 0x76`b4 DEC: 118`b4	+
3-bit LUT3/DFF/ LATCH6 -> OUT	NET19	HEX: 0x76`b5 DEC: 118`b5	+
3-bit LUT4/DFF/ LATCH7 -> Q	NET29	HEX: 0x76`b6 DEC: 118`b6	+
3-bit LUT5/DFF/ LATCH8 -> Q	NET30	HEX: 0x76`b7 DEC: 118`b7	+
3-bit LUT6/Pipe Delay/Ripple Counter -> OUT		HEX: 0x77`b0 DEC: 119`b0	-
3-bit LUT6/Pipe Delay/Ripple Counter -> Q1		HEX: 0x77`b1 DEC: 119`b1	-
3-bit LUT6/Pipe Delay/Ripple Counter -> Q2		HEX: 0x77`b2 DEC: 119`b2	-
FILTER/EDGE DET -> OUT		HEX: 0x77`b3 DEC: 119`b3	-
P DLY -> OUT		HEX: 0x77`b4 DEC: 119`b4	-
8-bit CNT1/DLY1 (MF1) -> OUT		HEX: 0x77`b5 DEC: 119`b5	-
OSC1 -> OUT0		HEX: 0x77`b6 DEC: 119`b6	-
OSC0 -> OUT0		HEX: 0x77`b7 DEC: 119`b7	-
OSC2 -> OUT		HEX: 0x78`b0 DEC: 120`b0	-
8-bit CNT2/DLY2 (MF2) -> OUT		HEX: 0x78`b1 DEC: 120`b1	-

Probe pins			
8-bit CNT3/DLY3 (MF3) -> OUT		HEX: 0x78`b2 DEC: 120`b2	-
8-bit CNT4/DLY4 (MF4) -> OUT		HEX: 0x78`b3 DEC: 120`b3	-
8-bit CNT5/DLY5 (MF5) -> OUT		HEX: 0x78`b4 DEC: 120`b4	-
8-bit CNT6/DLY6 (MF6) -> OUT		HEX: 0x78`b5 DEC: 120`b5	-
8-bit CNT7/DLY7 (MF7) -> OUT		HEX: 0x78`b6 DEC: 120`b6	-
4-bit LUT0 (MF0) -> OUT	NET27	HEX: 0x78`b7 DEC: 120`b7	+
3-bit LUT7 (MF1) -> OUT		HEX: 0x79`b0 DEC: 121`b0	-
3-bit LUT8 (MF2) -> OUT		HEX: 0x79`b1 DEC: 121`b1	-
3-bit LUT9 (MF3) -> OUT		HEX: 0x79`b2 DEC: 121`b2	-
3-bit LUT10 (MF4) -> OUT		HEX: 0x79`b3 DEC: 121`b3	-
3-bit LUT11 (MF5) -> OUT		HEX: 0x79`b4 DEC: 121`b4	-
3-bit LUT12 (MF6) -> OUT		HEX: 0x79`b5 DEC: 121`b5	-
3-bit LUT13 (MF7) -> OUT		HEX: 0x79`b6 DEC: 121`b6	-
16-bit CNT0/DLY0/ FSM0 (MF0) -> OUT		HEX: 0x79`b7 DEC: 121`b7	-
I2C -> OUT7		HEX: 0x7A`b0 DEC: 122`b0	-
I2C -> OUT6		HEX: 0x7A`b1 DEC: 122`b1	-
I2C -> OUT5		HEX: 0x7A`b2 DEC: 122`b2	-
I2C -> OUT4		HEX: 0x7A`b3 DEC: 122`b3	-

Probe pins

I2C -> OUT3		HEX: 0x7A`b4 DEC: 122`b4	-
I2C -> OUT2		HEX: 0x7A`b5 DEC: 122`b5	-
I2C -> OUT1		HEX: 0x7A`b6 DEC: 122`b6	-
I2C -> OUT0		HEX: 0x7A`b7 DEC: 122`b7	-
A CMP0H -> OUT	NET14	HEX: 0x7B`b0 DEC: 123`b0	+
A CMP1H -> OUT	NET15	HEX: 0x7B`b1 DEC: 123`b1	+
A CMP2L -> OUT	NET13	HEX: 0x7B`b2 DEC: 123`b2	+
A CMP3L -> OUT		HEX: 0x7B`b3 DEC: 123`b3	-
OSC1 -> OUT1		HEX: 0x7B`b4 DEC: 123`b4	-
OSC0 -> OUT1		HEX: 0x7B`b5 DEC: 123`b5	-
POR -> OUT	POR	HEX: 0x7B`b6 DEC: 123`b6	+
VDD (PIN 20) -> OUT	VDD	HEX: 0x7B`b7 DEC: 123`b7	+

Macrocells

<i>Macrocell</i>	<i>Property</i>	<i>Register</i>
16-bit CNT0/DLY0/ FSM0 (MF0)	Counted Data	HEX: 0x7C, 0x7D DEC: 124, 125
8-bit CNT2/DLY2 (MF2)	Counted Data	HEX: 0x7E DEC: 126
8-bit CNT4/DLY4 (MF4)	Counted Data	HEX: 0x7F DEC: 127